BEAM DESIGN SUMMARY

Project Name	:	G+2 RB BEAM DESIGN @FLOOR 2
Client Name	:	Unassigned
Engineer Name	:	Unassigned
Design File	:	E:\MY PROJECT 4-1\STAAD\RCDC FILES\G+2 Project Plan-Beam-1-8.077 m.R1.rcdx
Analysis File	:	E:\MY PROJECT 4-1\STAAD\G+2 Project Plan.std
Analysis Last Modified	:	06-10-2026 00:54:36
Level Designed	:	8.077 m

Beam No		:	**B1**	
Group No		:	**G1**	
Analysis Reference(Member)	8.077 m	:	5	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	56.72	47.65	23.39	61.23	23.5	81.94
PtClc (%)	0.414	0.342	0.2	0.451	0.2	0.632
AstCalc (sqmm)	352.66	291.15	170.2	384.17	170.2	538.07
Ast Prv (sqmm)	603.18	603.18	603.18	603.18	402.12	603.18
Reinforcement	3-T16	3-T16	3-T16	3-T16	2-T16	3-T16

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	53.23	50.05	73.61
Asv Torsion (sqmm/m)	-	171.95	245.67
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	718.14
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 140

SFR : -

Beam No		:	**B2**

Group No		:	**G1**	
Analysis Reference(Member)	8.077 m	:	6	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	31.4	32.21	40.21	65.02	30.94	83.48
PtClc (%)	0.219	0.225	0.285	0.483	0.216	0.647
AstCalc (sqmm)	186.35	191.48	242.41	411.14	183.48	550.19
Ast Prv (sqmm)	603.18	603.18	603.18	603.18	402.12	603.18
Reinforcement	3-T16	3-T16	3-T16	3-T16	2-T16	3-T16

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	62.39	38.97	62.52
Asv Torsion (sqmm/m)	209.46	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B3**	
Group No		:	**G1**	
Analysis Reference(Member)	8.077 m	:	141	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	20.46	20.46	20.46	55.33	45.76	29.27
PtClc (%)	0.2	0.2	0.2	0.403	0.327	0.203
AstCalc (sqmm)	170.2	170.2	170.2	343.11	278.65	173.12
Ast Prv (sqmm)	603.18	603.18	603.18	603.18	603.18	603.18
Reinforcement	3-T16	3-T16	3-T16	3-T16	3-T16	3-T16

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	113.51	110.08	106.65
Asv Torsion (sqmm/m)	707.17	696.43	685.69
Asv Reqd (sqmm/m)	491.56	465.87	440.18
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B4**	
Group No		:	**G2**	
Analysis Reference(Member)	8.077 m	:	25	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	68.71	36.85	33.96	59.33	28.38	73.79
PtClc (%)	0.515	0.259	0.238	0.436	0.2	0.559
AstCalc (sqmm)	437.89	220.8	202.45	370.85	170.2	475.57
Ast Prv (sqmm)	603.18	603.18	603.18	603.18	402.12	603.18
Reinforcement	3-T16	3-T16	3-T16	3-T16	2-T16	3-T16

Shear Design

	Left	Mid	Right
Vut (kN)	39.84	43.51	55.29
Asv Torsion (sqmm/m)	130.31	142.36	179.22
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B5**	
Group No		:	**G2**	
Analysis Reference(Member)	8.077 m	:	26	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	42.75	28.49	44.91	55.76	42.86	87.42
PtClc (%)	0.304	0.2	0.321	0.407	0.305	0.683
AstCalc (sqmm)	258.89	170.2	273.05	346.1	259.59	581.65
Ast Prv (sqmm)	603.18	603.18	603.18	603.18	402.12	603.18
Reinforcement	3-T16	3-T16	3-T16	3-T16	2-T16	3-T16

Shear Design

	Left	Mid	Right
Vut (kN)	41.55	36.5	52.96
Asv Torsion (sqmm/m)	134.46	119.99	171.21
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No : **B6**

Group No		:	**G2**	
Analysis Reference(Member)	8.077 m	:	146	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	18.68	18.68	21.51	78.5	59.74	29.9
PtClc (%)	0.2	0.2	0.2	0.601	0.439	0.208
AstCalc (sqmm)	170.2	170.2	170.2	511.3	373.72	177
Ast Prv (sqmm)	603.18	603.18	603.18	603.18	603.18	603.18
Reinforcement	3-T16	3-T16	3-T16	3-T16	3-T16	3-T16

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	75.82	74.99	72.41
Asv Torsion (sqmm/m)	308.88	306.29	298.21
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B7**	
Group No		:	**G3**	
Analysis Reference(Member)	8.077 m	:	45	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	67.63	35.44	38.94	61.17	28.62	75.14
PtClc (%)	0.505	0.249	0.275	0.451	0.2	0.571
AstCalc (sqmm)	430.04	211.84	234.2	383.8	170.2	485.72
Ast Prv (sqmm)	603.18	603.18	603.18	628.32	628.32	628.32
Reinforcement	3-T16	3-T16	3-T16	2-T20	2-T20	2-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	44.54	42.59	56.53
Asv Torsion (sqmm/m)	152.55	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		**:**	**B8**	
Group No		**:**	**G3**	
Analysis Reference(Member)	8.077 m	:	46	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	48.68	28.63	46.47	59.1	44.75	91.78
PtClc (%)	0.35	0.2	0.333	0.434	0.32	0.726
AstCalc (sqmm)	298.02	170.2	283.32	369.23	271.97	617.47
Ast Prv (sqmm)	603.18	603.18	603.18	628.32	628.32	628.32
Reinforcement	3-T16	3-T16	3-T16	2-T20	2-T20	2-T20

Shear Design

	Left	Mid	Right
Vut (kN)	43.11	41.52	57.13
Asv Torsion (sqmm/m)	142.89	148.57	198.46
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B9**	
Group No		:	**G3**	
Analysis Reference(Member)	8.077 m	:	151	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	21.41	21.41	21.41	72.08	55.49	32.53
PtClc (%)	0.2	0.2	0.2	0.544	0.404	0.227
AstCalc (sqmm)	170.2	170.2	170.2	462.73	344.2	193.47
Ast Prv (sqmm)	603.18	603.18	603.18	628.32	628.32	628.32
Reinforcement	3-T16	3-T16	3-T16	2-T20	2-T20	2-T20

Shear Design

	Left	Mid	Right
Vut (kN)	80.19	79.36	76.78
Asv Torsion (sqmm/m)	388.8	386.18	378
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No : **B10**

Group No		:	**G4**	
Analysis Reference(Member)	8.077 m	:	65	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	64.04	50.75	36.01	71.52	27.99	89.49
PtClc (%)	0.475	0.367	0.253	0.539	0.2	0.703
AstCalc (sqmm)	404.17	311.91	215.46	458.59	170.2	598.55
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	62.81	51.49	75.04
Asv Torsion (sqmm/m)	216.16	167.63	242.19
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 140

SFR : -

Beam No		:	**B11**	
Group No		:	**G4**	
Analysis Reference(Member)	8.077 m	:	66	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	43.6	31.93	43.88	74.17	39.36	98.93
PtClc (%)	0.311	0.223	0.313	0.562	0.278	0.797
AstCalc (sqmm)	264.4	189.7	266.25	478.37	236.89	678.35
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	62.85	47.41	70.96
Asv Torsion (sqmm/m)	203.23	156.19	230.75
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		**:**	**B12**	
Group No		**:**	**G4**	
Analysis Reference(Member)	8.077 m	:	156	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	11.01	11.01	15	65.53	48.3	18.83
PtClc (%)	0.2	0.2	0.2	0.487	0.347	0.2
AstCalc (sqmm)	170.2	170.2	170.2	414.86	295.52	170.2
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	60.39	74.95	73.75
Asv Torsion (sqmm/m)	220.81	354.43	350.61
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B13**	
Group No		:	**G5**	
Analysis Reference(Member)	8.077 m	:	134	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	18.26	22.07	14.08	10.37	4.85	14.52
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
Ast Prv (sqmm)	603.18	603.18	603.18	226.2	226.2	226.2
Reinforcement	3-T16	3-T16	3-T16	2-T12	2-T12	2-T12

Shear Design

	Left	Mid	Right
Vut (kN)	39.48	23.37	43.66
Asv Torsion (sqmm/m)	206.94	142.24	220.02
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No : **B14**

Group No		:	**G5**	
Analysis Reference(Member)	8.077 m	:	135	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	8.18	13.89	10.39	12.71	10.28	26.37
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
Ast Prv (sqmm)	603.18	603.18	603.18	226.2	226.2	226.2
Reinforcement	3-T16	3-T16	3-T16	2-T12	2-T12	2-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	30.47	20.66	35.32
Asv Torsion (sqmm/m)	142.56	119.85	157.72
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B15**	
Group No		:	**G5**	
Analysis Reference(Member)	8.077 m	:	159	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	16.07	16.07	16.07	34.57	30.51	25.23
PtClc (%)	0.2	0.2	0.2	0.242	0.213	0.2
AstCalc (sqmm)	170.2	170.2	170.2	206.3	180.84	170.2
Ast Prv (sqmm)	603.18	603.18	603.18	226.2	226.2	226.2
Reinforcement	3-T16	3-T16	3-T16	2-T12	2-T12	2-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	75.95	72.52	69.09
Asv Torsion (sqmm/m)	496.74	486	475.26
Asv Reqd (sqmm/m)	329.93	304.24	278.56
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B16**	
Group No		:	**G6**	
Analysis Reference(Member)	8.077 m	:	126	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	12.48	12.48	12.48	23.7	51.41	69.49
PtClc (%)	0.2	0.2	0.2	0.2	0.372	0.521
AstCalc (sqmm)	170.2	170.2	170.2	170.2	316.36	443.67
Ast Prv (sqmm)	603.18	603.18	603.18	603.18	603.18	603.18
Reinforcement	3-T16	3-T16	3-T16	3-T16	3-T16	3-T16

Shear Design

	Left	Mid	Right
Vut (kN)	68.74	75.51	75.51
Asv Torsion (sqmm/m)	407.77	428.95	428.95
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B17**	
Group No		:	**G6**	
Analysis Reference(Member)	8.077 m	:	111	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	25.1	28.27	28.55	81.88	22.84	66.24
PtClc (%)	0.2	0.2	0.2	0.632	0.2	0.494
AstCalc (sqmm)	170.2	170.2	170.2	537.52	170.2	419.99
Ast Prv (sqmm)	603.18	603.18	603.18	603.18	603.18	603.18
Reinforcement	3-T16	3-T16	3-T16	3-T16	3-T16	3-T16

Shear Design

	Left	Mid	Right
Vut (kN)	65.25	38.27	61.22
Asv Torsion (sqmm/m)	215.04	130.58	210.86
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No : **B18**

Group No		:	**G6**	
Analysis Reference(Member)	8.077 m	:	99	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	17.45	28.87	25.39	86.2	19.19	66.69
PtClc (%)	0.2	0.201	0.2	0.672	0.2	0.497
AstCalc (sqmm)	170.2	170.63	170.2	571.85	170.2	423.22
Ast Prv (sqmm)	603.18	603.18	603.18	603.18	603.18	603.18
Reinforcement	3-T16	3-T16	3-T16	3-T16	3-T16	3-T16

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	65.51	35.09	56.73
Asv Torsion (sqmm/m)	209.97	114.78	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B19**	
Group No		:	**G6**	
Analysis Reference(Member)	8.077 m	:	87	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	18.03	36.29	34.36	88.69	19.74	65
PtClc (%)	0.2	0.255	0.241	0.696	0.2	0.483
AstCalc (sqmm)	170.2	217.23	205	592.02	170.2	411.01
Ast Prv (sqmm)	603.18	603.18	603.18	603.18	603.18	603.18
Reinforcement	3-T16	3-T16	3-T16	3-T16	3-T16	3-T16

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	71.05	41.09	58.02
Asv Torsion (sqmm/m)	241.27	147.51	189.72
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		**:**	**B20**	
Group No		**:**	**G7**	
Analysis Reference(Member)	8.077 m	:	127	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	10.97	8.61	8.61	21.59	69.74	97.83
PtClc (%)	0.2	0.2	0.2	0.2	0.523	0.786
AstCalc (sqmm)	170.2	170.2	170.2	170.2	445.46	668.74
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	44.84	51.09	53.09
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B21**	
Group No		:	**G7**	
Analysis Reference(Member)	8.077 m	:	112	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	Mid	Right	Left	Mid	Right
Mud (kNm)	43.44	33.45	51.61	84.32	32.03	62.49
PtClc (%)	0.31	0.234	0.373	0.654	0.224	0.462
AstCalc (sqmm)	263.42	199.22	317.7	556.86	190.34	393.11
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	57.24	39.95	46.09
Asv Torsion (sqmm/m)	185	130.26	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No	:	**B22**
Group No	:	**G7**

Analysis Reference(Member)	8.077 m	:	100	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	29.56	29.96	40.74	83.6	28.67	54.45
PtClc (%)	0.206	0.208	0.289	0.648	0.2	0.396
AstCalc (sqmm)	174.89	177.42	245.79	551.11	170.2	337.04
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	54.29	35.05	42.36
Asv Torsion (sqmm/m)	177.34	116.43	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		**:**	**B23**	
Group No		**:**	**G7**	
Analysis Reference(Member)	8.077 m	:	88	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	26.69	36.85	47.82	80.33	25.64	52.06
PtClc (%)	0.2	0.259	0.343	0.617	0.2	0.377
AstCalc (sqmm)	170.2	220.79	292.29	525.44	170.2	320.76
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	54.89	35.83	41.14
Asv Torsion (sqmm/m)	178.57	118.24	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B24**	
Group No		:	**G8**	
Analysis Reference(Member)	8.077 m	:	128	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	19.96	11.21	11.21	21.25	63.78	88.93
PtClc (%)	0.2	0.2	0.2	0.2	0.473	0.698
AstCalc (sqmm)	170.2	170.2	170.2	170.2	402.32	593.96
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	83.02	86.51	72.53

Asv Torsion (sqmm/m)	465.69	476.73	339.54
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B25**	
Group No		:	**G8**	
Analysis Reference(Member)	8.077 m	:	113	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	36.06	33.56	40.24	96.82	31.07	79.84
PtClc (%)	0.254	0.235	0.285	0.776	0.217	0.613
AstCalc (sqmm)	215.77	199.93	242.54	660.03	184.33	521.63
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	71.44	44.46	68.42
Asv Torsion (sqmm/m)	238.58	153.16	243.59
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B26**
Group No		:	**G8**
Analysis Reference(Member)	8.077 m	:	101

Breadth	:	230	mm
Depth	:	430	mm
Concrete Grade	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover	:	30	mm
Side Clear Cover	:	30	mm
Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type	:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	25.16	35.84	34.76	99.04	25.61	76.83
PtClc (%)	0.2	0.252	0.244	0.798	0.2	0.586
AstCalc (sqmm)	170.2	214.35	207.51	679.32	170.2	498.51
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	72.06	41.53	60.86
Asv Torsion (sqmm/m)	238.7	142.75	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		**:**	**B27**
Group No		**:**	**G8**
Analysis Reference(Member)	8.077 m	:	89
Breadth		:	230 mm
Depth		:	430 mm
Concrete Grade		:	M25 N/sqmm
Grade Of Steel (Main)		:	Fe550 N/sqmm
Grade Of Steel (Shear)		:	Fe550 N/sqmm
Top/Bottom Clear Cover		:	30 mm
Side Clear Cover		:	30 mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016
Beam Type		:	Regular Beam

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	27.19	43.49	43.47	100.29	25.59	80.13
PtClc (%)	0.2	0.31	0.31	0.811	0.2	0.616
AstCalc (sqmm)	170.2	263.7	263.6	690.22	170.2	523.93
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	78.9	48.94	69.63
Asv Torsion (sqmm/m)	284.13	189.3	257.25
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B28**	
Group No		:	**G9**	
Analysis Reference(Member)	8.077 m	:	165	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	13.41	13.41	13.41	23.18	36.99	47.61
PtClc (%)	0.2	0.2	0.2	0.2	0.261	0.342
AstCalc (sqmm)	170.2	170.2	170.2	170.2	221.72	290.92
Ast Prv (sqmm)	603.18	603.18	603.18	339.3	339.3	339.3
Reinforcement	3-T16	3-T16	3-T16	3-T12	3-T12	3-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	62.58	69.19	72.26

Asv Torsion (sqmm/m)	416.8	437.49	447.09
Asv Reqd (sqmm/m)	254.81	254.81	257.32
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B29**	
Group No		:	**G9**	
Analysis Reference(Member)	8.077 m	:	166	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	9.59	10.39	2.66	39.05	16.32	19.97
PtClc (%)	0.2	0.2	0.2	0.276	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	234.87	170.2	170.2
Ast Prv (sqmm)	603.18	603.18	603.18	339.3	339.3	339.3
Reinforcement	3-T16	3-T16	3-T16	3-T12	3-T12	3-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	33.37	14.78	30.36
Asv Torsion (sqmm/m)	134.4	54.25	140.82
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B30**
Group No		:	**G9**
Analysis Reference(Member)	8.077 m	:	167

Breadth	:	230	mm
Depth	:	430	mm
Concrete Grade	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover	:	30	mm
Side Clear Cover	:	30	mm
Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type	:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	5.5	14.23	7.41	22.04	1.14	20.43
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
Ast Prv (sqmm)	603.18	603.18	603.18	339.3	339.3	339.3
Reinforcement	3-T16	3-T16	3-T16	3-T12	3-T12	3-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	29.64	12.81	29
Asv Torsion (sqmm/m)	107.54	54.87	105.55
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		**:**	**B31**
Group No		**:**	**G9**
Analysis Reference(Member)	8.077 m	:	168
Breadth		:	230 mm
Depth		:	430 mm
Concrete Grade		:	M25 N/sqmm
Grade Of Steel (Main)		:	Fe550 N/sqmm
Grade Of Steel (Shear)		:	Fe550 N/sqmm
Top/Bottom Clear Cover		:	30 mm
Side Clear Cover		:	30 mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016
Beam Type		:	Regular Beam

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	11.52	24.83	21.14	26.62	4.58	15.31
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
Ast Prv (sqmm)	603.18	603.18	603.18	339.3	339.3	339.3
Reinforcement	3-T16	3-T16	3-T16	3-T12	3-T12	3-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	47.85	31.29	40.66
Asv Torsion (sqmm/m)	228.58	176.75	206.06
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -